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A Multi-Level Clustering Approach for Forecasting Taxi Travel Demand

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Abstract— **In this paper, we use time-series modeling to forecast taxi travel demand, in the context of a mobile application-based taxi hailing service. In particular, we model the passenger demand density at various locations in the city of Bengaluru, India. Using the data, we first shortlist time-series models that suit our application. We then analyse the performance of these models by using Mean Absolute Percentage Error (MAPE) as the performance metric. In order to improve the model performance, we employ a multi-level clustering technique where we aggregate demand over neighboring cells/geohashes. We observe that the improved model based on clustering leads to a forecast accuracy of 80% per km^2. In addition, our technique obtains an accuracy of 89% per km^2 for the most frequently occurring use case.**

Keywords: Travel demand forecasting, Multi-level clustering, Time-series, Transportation networks, Data analysis.

I. INTRODUCTION

Transportation planning is a collaborative process involving public and private agencies. Public transport remains the primary mode of transport for most of the population, and India's public transport system is among the most heavily used in the world. However, this sector has not been able to keep up with the rising demand [12]. Hence, to match transportation demand, an alternate mode of transport such as taxi services are essential. This sector has undergone drastic changes with the introduction of mobile application based taxi hailing services. From the GPS data generated by these apps, the demand (number of taxi booking requests) generated by passengers can be obtained. In this paper, we tackle the problem of analysing and modeling this taxi travel demand, using a multi-level clustering technique based on time-series.

A time-series model is built on the assumption that the present and the future demand would have some correlation with the past demand, and can be represented as the function of the past data. Time-series analysis can provide accurate results for a short-term, but tends to flatten out in the long-term [6]. Therefore, it is suitable for short-term forecasting of demand and supply. Indeed, accurate forecasting of taxi demand and supply is crucial for mitigating demand-supply imbalance, by routing idle drivers to passenger hotspots. In particular, it is important to understand the dynamics of travel demand. To that end, we model the demand by employing time-series techniques on a real-world dataset containing the taxi demand generated for the city of Bengaluru, India. This dataset was acquired from a leading app-based taxi hailing service provider in India. Using these time-series models, we predict future demand to reduce the demand-supply imbalance.

Traffic analysis has been an active research area over the recent years. A fundamental approach for traffic analysis is to understand human mobility, *i.e*, draw statistical inferences from empirical data [3]. The spatio-temporal regularity of human mobility is measured in [5]. We are particularly concerned with taxi demand analysis; a couple of related works are [8], [12]. A study in Beijing [8], suggests that human mobility has bursty characteristics and taxi's traveling displacements tend to follow an exponential distribution rather than power law. A simultaneous equation system of passenger demand, taxi utilization and level of services that can be used for fixing the taxi fare structure is developed in [13].

Previous studies on taxi model fitting and prediction mainly deal with predicting distribution of vacant taxis [10], recommender systems for taxi drivers and passengers based on the hotspots [14], and predicting human mobility from passenger pickups [7]. The work in [7] is closely related to ours – they predict the availability of pickups (similar to demand generated in our case) in a hotspot in the next time interval for China and specifically, an improved ARIMA model was observed as the best fit for their dataset. Empirically, we noticed that Indian traffic undergoes changes very rapidly from area to area, and hence a single unified model may not suffice. To the best of our knowledge, there has been little research on the Indian taxi demand modeling using data analysis. This is the focus of the current work.

Each location in the city is identified by a unique set of alphanumeric characters, known as a geohash. A 6-level geohash has six alphanumeric characters and encloses an area of approximately 1.2 km $\times$ 0.6 km. In this paper, the demand density for the most heavily used geohashes in the city of Bengaluru, India is modeled and fitted using time-series decomposition analysis. First, we identify the models that best fit our data. Then, in order to improve the model performance, we propose a multi-level clustering technique that uses the correlation between adjacent geohashes to reduce the forecast error. The performance of the model improved by 20% when this technique was implemented. For a 6-level geohash, an accuracy of 89% was obtained for the range in which most of the demand falls.

The rest of the paper is organized as follows. In Section II we review the models that worked for our data, followed by a discussion on the model fitting and results in Section III. In Section IV the results from the improved model, incorporating neighboring cells information, are presented.

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In Section V, applications of our work are discussed. Finally, in Section VI, we summarize our contributions.

II. Time-series model selection

Of the various time-series models available in the literature, the models that are widely used are moving average, exponential smoothing, *i.e*, Holt-Winters (HW) model, Seasonal Naive, FFT, STL decomposition, ARIMA, TBATS, linear regression and state space ETS model [6]. In this paper, we restrict our attention to linear models. Since different geohashes have varying characteristics, a single model may not provide a good fit for all the geohashes. For example, the model for a geohash in residential area may not be a good fit for a geohash in a shopping area. For this reason, we use different time-series models for forecasting demand in different types of geohashes. Between the models that are mentioned below, almost all types of geohashes are well-modelled for our dataset.

Let y_t be the number of taxi bookings at time t.

- **Baseline model**: An averaging model is used as a baseline to compare different time-series models. A model is retained if it performs better than the baseline, else it is discarded. In this model, we perform averaging over all the past seasons; for example, if weekly periodicity is considered, the prediction for Monday at 9 AM is the average of all past Monday 9 AM demand. More generally, we have:

$$y_{t+1} = \frac{1}{N}\sum_{i=1}^{N} y_{t+1-im}, \tag{1}$$

  where N is the total number of seasons and m is the seasonality period.
- **Linear regression on trend and season**: The season and the trend are deduced by decomposition from the data provided. The time-series is then regressed on its trend and seasonal components.
- **Seasonal and Trend decomposition using Loess (STL)** [1]: In this method, a fit is achieved by first decomposing the data into a seasonal and a seasonally adjusted component. For the seasonal component S_t, we conduct a Seasonal Naive forecast since it is assumed that the seasonal component changes extremely slowly. For the seasonally adjusted component A_t, any of the common techniques like ARIMA, ETS, naive and random drift can be used. Out of these shortlisted techniques, STL had the best performance. The decomposition equations are:

$$y_t = S_t + A_t, \tag{2}$$

$$S_{T+h} = S_{T+h-km}; \quad k = \lfloor (h-1)/m \rfloor + 1. \tag{3}$$

  Here, $\lfloor u \rfloor$ denotes the integer part of u, h is the forecast horizon and m is the season.
- **TBATS model** [4]: TBATS model can deal with time-series having multiple seasonalities, high frequency seasonality, non integer seasonality and calendar effects. The identifier BATS is an acronym for key features of the model: Box-Cox transform [6], ARMA errors d_t, Trend b_t and Seasonal components $s_t^{(i)}$. The seasonal components are represented in terms of trigonometric Fourier series, and hence the first T (Trigonometric) in TBATS. The equations are outlined below:

$$y_t = \ell_{t-1} + \phi b_{t-1} + \sum_{i=1}^{T} s_{t-1}^{(i)} + d_t, \tag{4}$$

$$\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha d_t, \tag{5}$$

$$b_t = (1-\phi)b + \phi b_{t-1} + \beta\ d_t, \tag{6}$$

$$d_t = \sum_{i=1}^{p} \varphi_i d_{t-i} + \sum_{i=1}^{q} \theta_i \varepsilon_{t-i} + \varepsilon_t, \tag{7}$$

$$s_t^{(i)} = \sum_{j=1}^{k_i} s_{j,t}^{(i)}, \tag{8}$$

  where ℓ_t is the local level in period t, b is the long run trend, b_t is the short run trend, d_t is an ARMA(p,q) process, ε_t~N(0, σ^2), T seasonal patterns, α and β are the smoothing parameters, $s_t^{(i)}$ is the ith seasonal component at time t, $s_{j,t}^{(i)}$ is it's j^{th} harmonic component and k_i are the number of harmonics needed to represent $s_t^{(i)}$.
- **Exponential smoothing / Holt Winters (HW) model** [11]: To produce a forecast, HW needs to estimate three components, namely, the level ℓ_t, the trend b_t and the season s_t. Level is the underlying constant demand that remains after de-seasonalising the data and removing the effect of random factors (noise). Trend refers to the change in that underlying level over time. Season is the frequency with which a certain behavior of the data repeats. We used HW model with weekly and daily seasonality. The weight attached with the data decreases exponentially, as the data gets older, as revealed from the equations below:

$$y_{t+h|t} = \ell_t + hb_t + s_{t-m+h_m^+}, \tag{9}$$

$$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}), \tag{10}$$

$$b_t = \beta\ (\ell_t - \ell_{t-1}) + (1-\beta)\ b_{t-1}, \tag{11}$$

$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m}, \tag{12}$$

  where α, β and γ are the smoothing parameters, m is the seasonal period and $h_m^+ = \lfloor (h-1) \mod m \rfloor + 1$.

A. Metric for Performance Evaluation

For evaluating the performance of a specific model, we consider Mean Absolute Percentage Error (MAPE) as the metric, which is calculated as:

$$M = \frac{1}{N}\sum_{t=1}^{N}\left|\frac{A_t - F_t}{A_t}\right|, \tag{13}$$

where N is the number of samples, A_t is the actual value and F_t is the forecast at time t. MAPE is chosen as our preferred metric because it puts a heavier penalty on negative errors (actual $<$ forecast) than on positive errors. Penalizing negative errors works in our favor because if the forecast

demand exceeds the actual demand, we may wastefully route more drivers than necessary.

B. Model Validation

Checking the residuals from the fitting can help in further validation of the selected model. For this, we need to examine:

- Whether the probability distribution of errors follow a normal distribution with zero mean and constant variance.
- If errors satisfy the required p-value from a statistical test like Box-Ljung test, that supports the lack of correlation between errors.
- The correlation between residuals by consulting with its auto correlation plot, for further validation.

If the residuals sufficiently satisfy these conditions [6], it gives us confidence in the underlying model being suitable for the given time-series. The shortlisted models are those that gave sufficiently good results during residual analysis and also resulted in low MAPE. Once the models are shortlisted, we fit the data after pre-processing and compare different models, as elaborated in the next section.

III. Modeling Demand

Most of the previous work with demand modeling was limited to a short-duration [2] [9], because of which the key characteristics of data might not be captured. We worked with a rich dataset spanning over two months, with nearly 7 million taxi trips each month, which gives us more confidence in capturing the level, trend and seasonal components.

The data was acquired from a leading Indian transportation company dealing with app-based taxi rental services. It contains the taxi booking demand generated by the public, along with user identification numbers, location in the form of geohash and time. Hourly demand is defined as the count of passengers opting for a taxi in an hour. The goal of our work is to model and predict hourly demand generated at a 6-level geohash, so that taxi allocation can be performed every hour. Every city can be divided into geohashes for the ease of allocating taxis and locating demand. A 6-level geohash has six alphanumeric characters and encloses an area of approximately 1.2 km $\times$ 0.6 km. For a lower level geohash, namely a 5-level geohash, the area covered is more than a 6-level geohash and therefore, the precision reduces.

A. Data Pre-processing

Some of the techniques we used for pre-processing our data are outlined here. A customer can access the taxi rental mobile application to avail booking/tracking taxi facility. The first use case is considered as possible demand. If a user ID is detected multiple times in an hour in the same geohash, it is taken as a single session because it is unlikely that a customer will make multiple bookings in the same hour, from the same geohash. Since the traffic conditions change very rapidly, we need to forecast almost in real-time. Linear time-series models are amenable for real-time forecasting because they are computationally effecient. In order to perform linear modeling, a Box-Cox transformation [6], a type of logarithmic transformation, is first applied to the raw data. This stabilizes the variance of the data.

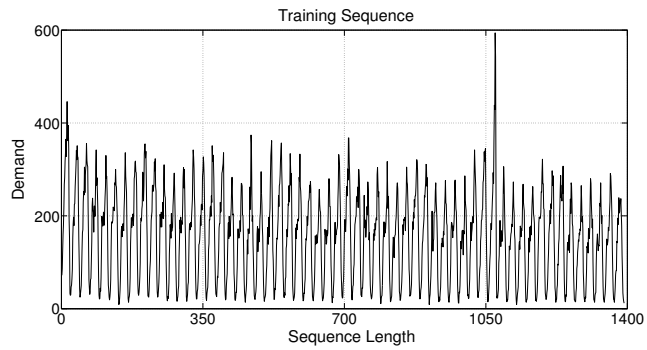


Fig. 1: A typical time-series used for training. The plotted time-series is for one geohash spanning over 59 days.

TABLE I: Training Sequence Characteristics

Characteristic	Value
Minimum	9
Mean	155.26
Maximum	594
Skewness	0.25
Kurtosis	- 0.44
Standard Deviation	95.02
Inter-Quartile Range	158
Sequence Length	1392 samples
Seasonal Periods	24 samples, 168 samples

B. Model Fitting

Once the data is pre-processed, the next step is to analyse it using the models mentioned in Section II. To see where the demand is mostly concentrated, all the geohashes in the city are sorted in the decreasing order of passenger demand count and we concentrate on the top 20 percent of the geohashes in this category. A macroscopic view of the demand patterns in the city can be obtained by analysing the top geohashes. The same techniques can be extended to other geohashes.

For every geohash, an hourly time-series is formulated for 2 months. Every value in the time sequence is equivalent to an hour in time and represents the aggregate demand generated in the entire hour. We train our models on the training set and then use it on the test set to determine the model with the minimum error. Since we need to capture maximum characteristics of the data and predict for the next day, we use the first 59 days of time-series to train, and test it for the 60th day. The model with the least MAPE is selected for that particular geohash.

Figure 1 shows a typical training sequence. It's key statistical characteristics are listed in Table I. See Figures 2 and 3 for the prediction results on the top 2 geohashes.

C. Results from the Basic Model

Considering the best models for each geohash, we obtained an average MAPE of 25% for the top 20 percent

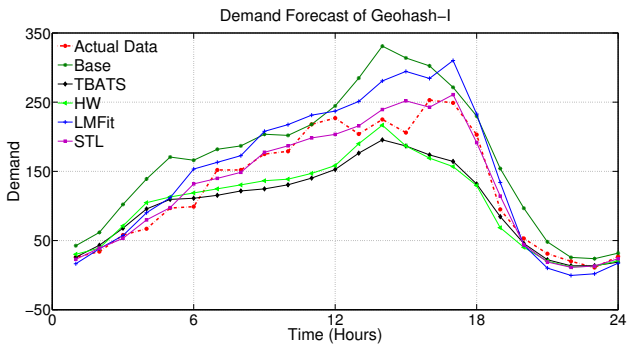


Fig. 2: Comparison of the actual test data with the forecasts of five shortlisted models over one day for Geohash-1.

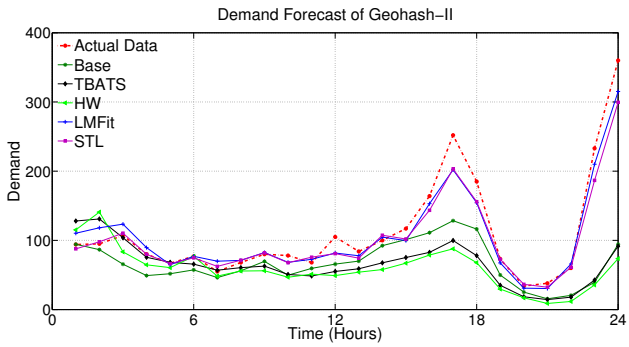


Fig. 3: Comparison of the actual test data with the forecasts of five shortlisted models over one day for Geohash-2.

geohashes. There are around 80 geohashes in the top 20 percent. Since this MAPE is quite high considering that the error was for the top geohashes, the natural next step is to find the MAPE at a lower level (5-level or a finer level) and use that information to improve the result.

The metric MAPE depends on the magnitude of the numerical values used in the computation. Since a 6-level geohash is approximately a 1 km^2 area, it tends to have low demand during early morning and late nights. It tampers with the MAPE calculation and drives MAPE high at some odd hours. Hence, in the next step, the 5-level geohashes that contain the top eighty 6-level geohashes were analysed. A 5-level geohash covers an area of 4.9 km $\times$ 4.9 km and has sufficient demand density at all hours. The average MAPE was observed to be 8%. It indicates that, at a 25 km^2 level, the hourly demand generated for a day can be predicted with an accuracy of 92%.

Delving deeper, we cluster 9 adjacent 6-level geohashes together. See Figure 4 for the schematic representation of clusters. Let $\mathcal{T} = \{T_i\} \; \forall \, i \in \{1, \ldots, 80\}$ be the set of top 80 geohashes. T_i is the ith top geohash. Define a function $\mathcal{N}(T_i)$ that generates the 8 immediate neighboring geohashes of the corresponding T_i. Then, a cluster R_i is defined as:

$$R_i = \mathcal{N}(T_i) \cup T_i. \tag{14}$$

$\mid R_i \mid$ is fixed as 9. Let $D_t(x)$ be a function that returns the aggregated demand in the region x at time t. Note that x can either be a geohash or a cluster of geohashes. Hence,

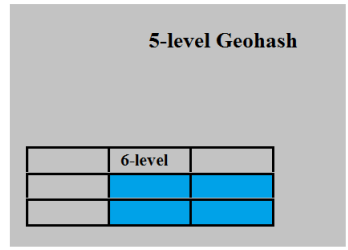


Fig. 4: The grey region represents a 5-level geohash and the smallest single cell inside it is a 6-level geohash (32 6-levels in a 5-level). The 9 cells inside the 5-level form a R cluster and the 4 blue cells together form a Q cluster inside R.

$D_t(R_i)$ returns the aggregate demand of cluster R_i at time t. This procedure is repeated over each cluster, per hour, to form a time-series. The average MAPE for these R clusters was calculated and it is approximately 10%. After further analysis, it became evident that travel demands across nearby geohashes are correlated and that the demand is scattered inside a cluster among several geohashes without a fixed pattern. This led to an increased MAPE.

In the next section, we propose a multi-level clustering technique that improves the model performance by exploiting the correlation between neighboring geohashes.

IV. Improving Accuracy by Multi-Level Clustering

For a more precise demand prediction, in addition to fitting a linear model to the data, we gather information from neighboring cells.

We propose two multi-level clustering approaches below:

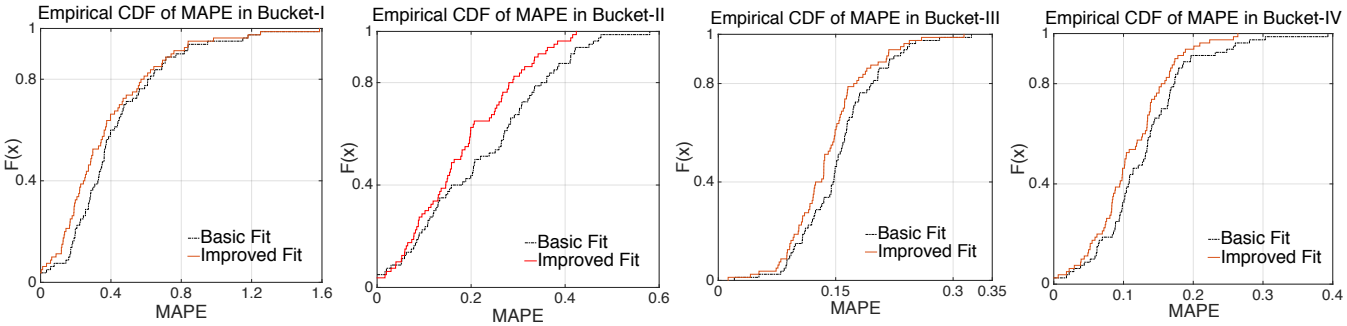
- *Approach 1*: The correlation between immediate neighbour geohashes motivates this approach. For every hour, we express the demand of the central geohash $D(T_i)$ as a fraction of its cluster R demand $D(R_i)$ and formulate a percentage time-series Γ_i. Let $r_j \in R_i$, where each r_j is a 6-level geohash. Let j = 1, . . ., 9. The fraction $\Gamma_{i,t}$ at time t is defined as:

$$\Gamma_{i,t} = \frac{D_t(T_i)}{\sum\limits_{r_j \in R_i} D_t(r_j)}. \tag{15}$$

  The fractions are predicted for the next day (test period) and is multiplied with the predicted cluster R demand (forecasted for the test period) to get the corresponding 6-level demand forecast $\hat{D}(T_i)$.

$$\hat{D}_t(T_i) = \hat{\Gamma}_{i,t} \cdot \hat{D}_t(R_i). \tag{16}$$

- *Approach 2*: Inside an R cluster, a group of geohashes contributed to most of the demand inside the cluster, and this pattern remained consistent over time. This consistency motivates *Approach 2*. In a cluster R_i, there can be more than one 6-level geohash r_j that belongs



(a) Bucket-I (B_1) (b) Bucket-II (B_2) (c) Bucket-III (B_3) (d) Bucket-IV (B_4)

Fig. 5: Comparison of MAPE before and after multi-level clustering, for the top 80 geohashes.

to T. Those $\{r_j\}$s are grouped together and named as sub-cluster Q_i.

$$Q_i = R_i \cap T. \tag{17}$$

Two levels of clustering and forecasting is performed here. First, we express the demand of the cluster Q $D(Q_i)$ as a fraction of $D(R_i)$ and formulate a fraction time-series Λ_i. At time t, the time-series Λ_i is defined as:

$$\Lambda_{i,t} = \frac{D_t(Q_i)}{\sum\limits_{r_j \in R_i} D_t(r_j)}. \tag{18}$$

The demand forecast for Q_i at time t is then calculated as:

$$\hat{D}_t(Q_i) = \hat{\Lambda}_{i,t} \cdot \hat{D}_t(R_i). \tag{19}$$

The average MAPE of Q clusters was found to be 12%. Then, a second level of clustering and prediction is done where $D(T_i)$ is represented as a fraction of $D(Q_i)$ and a fraction time-series Ψ_i is formulated:

$$\Psi_{i,t} = \frac{D_t(T_i)}{\sum\limits_{r_j \in Q_i} D_t(r_j)}. \tag{20}$$

Now, the forecast for the corresponding 6-level geohash can be obtained as:

$$\hat{D}_t(T_i) = \hat{\Psi}_{i,t} \cdot \hat{D}_t(Q_i). \tag{21}$$

In other words, instead of predicting demand directly from a 6-level geohash, it is predicted indirectly by expressing the 6-level demand as a fraction of R and Q clusters. Now, we have 6-level demand forecast from 3 techniques; $D^{(1)}(T_i)$ from basic model fitting, $D^{(2)}(T_i)$ from R fraction fitting and $D^{(3)}(T_i)$ from Q fraction fitting. Even though there are 2 levels of prediction in each technique now, since we use linear models, it does not vastly increase the time needed for computation. Table II contains MAPE at different geohash levels.

TABLE II: MAPE at different levels

Level	5-level	cluster R	cluster Q	6-level
MAPE (%)	8	10	12	20

A. Improvement in MAPE due to Multi-Level Clustering

Taking the best forecast $D^*(T_i)$ from the 3 techniques mentioned above, the average MAPE for the top 20 percent of geohashes reduced to 20%, *i.e.*, a 5% decrease in overall MAPE. In other words, the multi-level clustering technique led to a 20% improvement in the overall MAPE.

We observed that 75% of the hourly demand in the top 80 geohashes lies in the range 40-70 bookings. Motivated by this, we are particularly interested in ensuring a low MAPE for this range, where most of the demand is concentrated. In order to find the MAPE over different ranges of demand, we follow these steps:

1) Based on the magnitude of hourly demand, geohashes were divided into different sets, termed as *buckets*.
2) All the demand falling in 0-10 magnitude range was included in bucket B_1. Similarly, demand ranges were divided into 10-20, 20-50, 50-100 and above 100, and were assigned to buckets B_2, B_3, B_4 snd B_5 respectively.
3) The MAPE for each bucket was calculated separately.

We observe that the frequently occurring demand range (40-70) falls in buckets B_3 and B_4, and that the MAPE for these buckets are 14% and 11% respectively. This highlights that the model based on our proposed multi-level clustering technique succeeded in capturing the key characteristics of the data better than the basic model. We were able to achieve an accuracy of 89% for the most frequently occurring use case. Figure 5 compares MAPE between the basic model fitting and the improved model fitting based on multi-level clustering. The figure depicts the cumulative distribution (CDF) plots of MAPE for the top 80 geohashes, before and after clustering. We can observe that the CDF of the improved model covers more sample instances at low error magnitudes. This clearly shows the superiority of the improved model based on multi-level clustering over the basic model. Results for bucket B_5 are not plotted because only in a few top geohashes, the hourly demand shot above 100. Table III compares the MAPE before and after clustering for the top 30 6-level geohashes.

The immediate applications of our work are mentioned in the next section.

TABLE III: MAPE for top geohashes, averaged over all buckets, for Basic Model (M^B) and Improved Model (M^*)

Geohash	M^B	M^*	**Geohash**	M^B	M^*	**Geohash**	M^B	M^*
1	18.8	12.5	11	21.4	18.2	21	17.6	14.9
2	17.0	15.6	12	23.4	21.0	22	14.3	8.4
3	27.0	26.9	13	44.9	36.6	23	23.3	17.7
4	13.3	10.0	14	23.0	13.5	24	13.1	9.2
5	40.3	32.5	15	23.1	12.2	25	23.5	17.4
6	22.5	15.2	16	24.4	19.6	26	23.3	20.7
7	27.0	24.1	17	19.4	17.4	27	50.3	46.8
8	12.4	12.2	18	23.2	20.9	28	26.2	19.0
9	34.1	26.6	19	20.3	16.7	29	18.2	13.5
10	23.7	16.4	20	14.6	12.9	30	22.0	16.4

V. Application to Demand prediction

The taxi driver sends out information regarding his location every few seconds using Global Positioning System (GPS). This data can be filtered to find the supply, *i.e*, availability of taxis ready to take in customers, at different locations. If we project supply as a time-series per hour per geohash, time-series modeling can be applied there. Once we have both demand and supply data, we can aim to address the demand-supply imbalance problem. Another application of demand analysis is to detect passenger hotspots in the city.

Demand prediction is an integral part of idle taxi reallocation problem. If we know that the demand will be high in a region and the supply will be low in that region, then we can readily route more drivers to this high demand region. For example, for a 6 level geohash, we can accurately predict the demand and supply for the next time interval in that area. If the demand for taxi is predicted to increase and the supply is predicted to be low there, we find idle taxis (this information is obtained from the taxi pings given out during its course) within a few kilometer radius of this location, that can reach the location in a fixed time and route those taxis there.

VI. Conclusions

Travel demand modeling is an intrinsic part of transportation planning. Analysing demand can help us in finding passenger hotspots, balancing demand-supply problem and re-allocating taxis to aid drivers find customers.

In this paper, the passenger demand for taxis was modeled using time-series analysis. Time-series decomposition techniques were readily applied to the data and a couple of techniques that fit the Indian taxi demand data were shortlisted. A multi-level clustering technique was proposed to improve the accuracy of linear time-series model fitting, by exploring the correlation between adjacent geohashes. A 20% improvement in results was observed when the correlation information was incorporated into the model. We were able to achieve an accuracy of 89% at 1 km^2 area level, for the most common demand range.

A natural extension of our work would be to improve the accuracy/km^2. We employed linear time-series modeling with MAPE as the performance metric. Non-linear models along with other performance metrics should be explored in future studies.

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